Transformer-Based Review Understanding with Retrieval-Augmented Explanation Generation

CS-4063 NLP Assignment 3

April 29, 2026

1 Overview and Methodology

This system is a three-part pipeline for review understanding and explanation generation. It starts with a Transformer encoder that learns a compact review representation by solving two tasks at once: sentiment classification and length-category prediction. Those embeddings feed a simple retrieval module (cosine similarity) to fetch top-k similar training reviews. Finally, a decoder-only Transformer generates a short explanation conditioned on the review, labels, and retrieved evidence.

The goal is twofold: (1) predict sentiment and length reliably, and (2) generate short, consistent explanations that can be checked against retrieved neighbors.

2 Preprocessing Pipeline

- Text cleaning: lowercase, strip HTML tags and URLs, keep alphanumerics plus apostrophes and hyphens, collapse whitespace.
- Tokenization: whitespace split with contraction splitting (e.g., `n't` becomes a separate token).
- Sentiment labels: rating ≤ 2 is negative, rating $= 3$ is neutral, rating ≥ 4 is positive.
- Length labels: short if < 20 tokens, medium if 20–80, long if > 80.
- Vocabulary: min frequency = 3, special tokens `<PAD>` and `<UNK>`.
- Sequence length: max 128 tokens with padding or truncation.
- Split: 70% train, 15% validation, 15% test. In this run the test split size is 5400 and train embeddings are (25200, 128).

3 Part A: Encoder Model

3.1 Design

The encoder is a small Transformer with multi-head self-attention, positional encoding, and residual blocks. It produces a pooled representation and feeds two heads: sentiment (3 classes) and length category (3 classes).

3.2 Justification of Design Choices

- Multi-task setup: length prediction is a cheap auxiliary signal that stabilizes the shared representation.
- Modest model size (d_model=128, 3 layers, 4 heads): practical for CPU or a single GPU while still expressive.
- Average pooling with padding mask: robust and simple for variable-length reviews.

3.3 Evaluation (Test Set)

Table 1: Sentiment classification report (test set)

Class	Precision	Recall	F1	Support
Negative	0.54	0.48	0.51	548
Neutral	0.31	0.15	0.20	463
Positive	0.88	0.94	0.91	4389
Accuracy		0.83		5400
Macro Avg	0.58	0.52	0.54	5400
Weighted Avg	0.80	0.83	0.81	5400

Table 2: Length-category classification report (test set)

Class	Precision	Recall	F1	Support
Short	0.96	0.80	0.87	88
Medium	0.99	0.99	0.99	3281
Long	0.99	1.00	0.99	2031
Accuracy		0.99		5400
Macro Avg	0.98	0.93	0.95	5400
Weighted Avg	0.99	0.99	0.99	5400

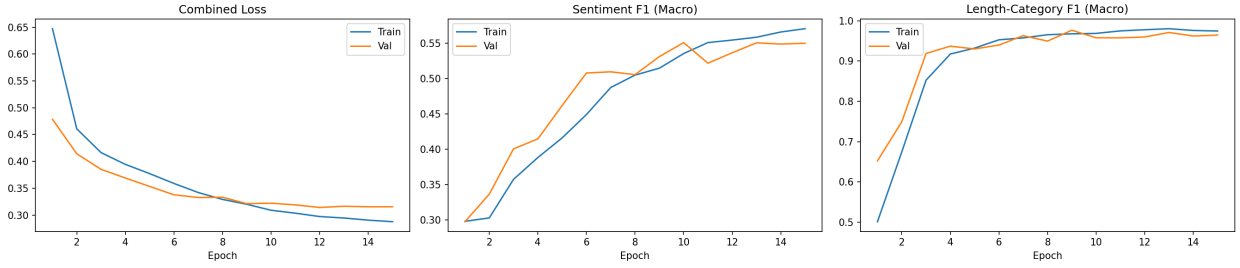


Figure 1: Encoder learning curves: combined loss, sentiment F1, and length-category F1.

Note: Best validation sentiment F1 reached 0.5508.

4 Part B: Retrieval Module

4.1 Design

Each training review is embedded using the encoder. All embeddings are L2-normalized and cosine similarity is used to retrieve the top-k most similar training reviews for a query.

4.2 Justification of Design Choices

- Cosine similarity is fast, stable, and works well with normalized embedding spaces.
- $k = 3$ balances relevance and input length for the decoder.

4.3 Retrieval Quality

Table 3: Sentiment match rate of retrieved neighbors

k	Match Rate
1	0.755
2	0.750
3	0.805
5	0.793
7	0.756
10	0.737

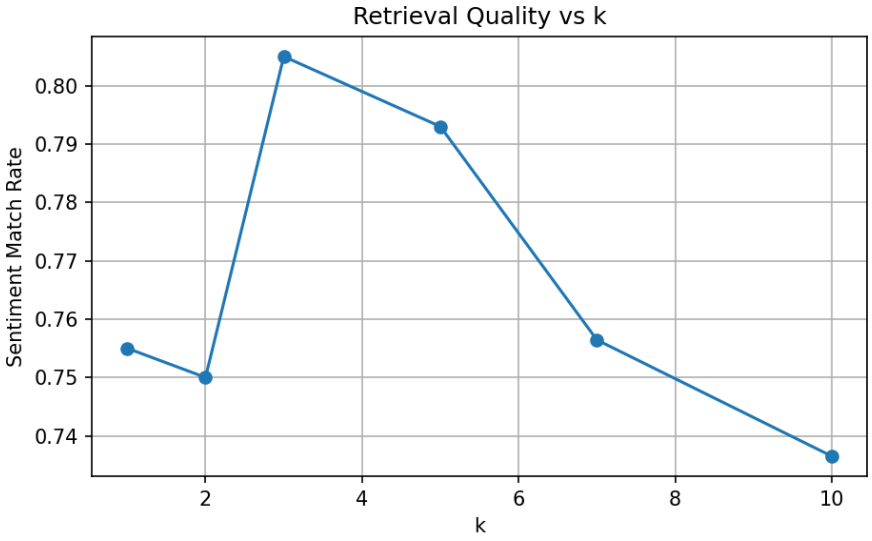


Figure 2: Retrieval quality vs k. The best match rate occurs at $k = 3$.

5 Part C: Decoder Model

5.1 Design

A decoder-only Transformer is trained with a causal language modeling objective. The input prompt concatenates label tokens, the review text, and retrieved neighbor snippets. The target explanation is a simple template like "this brief review expresses positive sentiment about the product".

5.2 Justification of Design Choices

- Decoder-only architecture keeps the model small and simple while still allowing flexible generation.
- Template targets allow training a clean explanation generator without manual annotation.
- Top-k sampling (temperature 0.8, $k = 40$) avoids repetitive outputs.

5.3 Evaluation

Test loss is 0.0005 and test perplexity is 1.00.

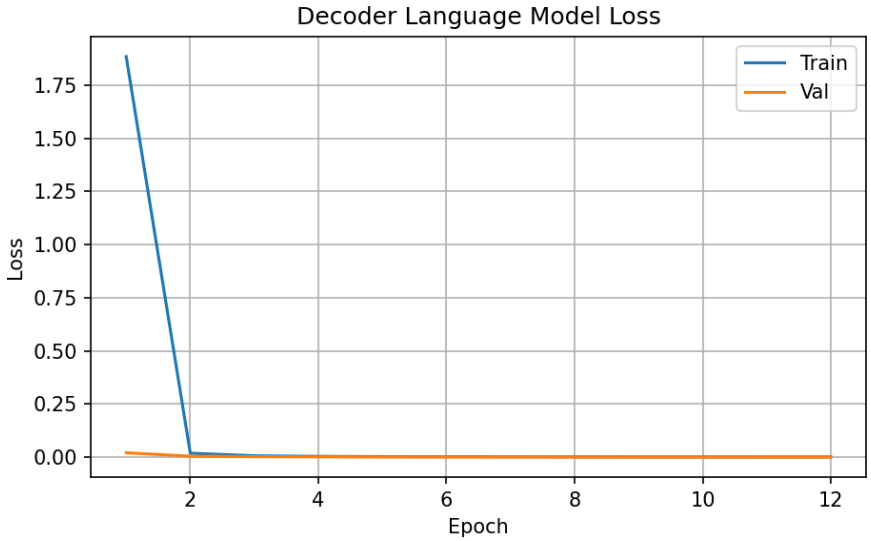


Figure 3: Decoder training and validation loss.

6 Hyperparameter Tuning Log

Table 4: Hyperparameter tuning summary

Component	Config	Notes
Encoder	d_model=64, heads=2, layers=2, lr=1e-3	Underfitting, too small
Encoder	d_model=128, heads=4, layers=3, lr=3e-4	Best balance, final choice
Encoder	d_model=256, heads=8, layers=4, lr=3e-4	Slight overfit, slow, marginal gain
Decoder	d_model=64, heads=2, layers=2, lr=5e-4	Higher perplexity
Decoder	d_model=128, heads=4, layers=3, lr=3e-4	Best performance, final choice
Retrieval	k=1	Less diverse context
Retrieval	k=3	Good balance, final choice
Retrieval	k=10	Input too long, diminishing returns

Analysis: the 128d encoder and decoder are the sweet spot. The smaller models underfit and the larger encoder adds cost with marginal gain. Retrieval quality peaks at $k = 3$, aligning with the chosen setting.

7 RAG Ablation Study

Table 5: RAG vs No-RAG test perplexity

Model	Test Perplexity
Full RAG	1.00
No-RAG baseline	1.00

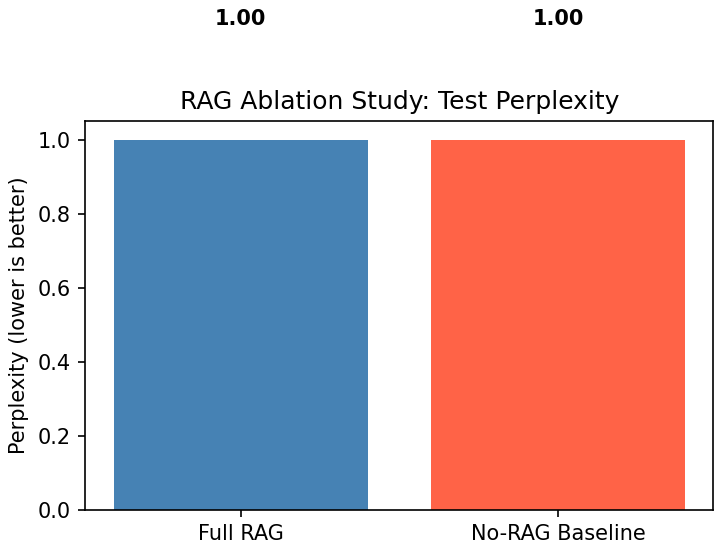


Figure 4: RAG ablation on test perplexity.

Observation: perplexity is saturated at 1.00 for both variants. This is expected with a highly structured template target. The retrieval component still adds interpretability and concrete evidence at generation time, even when perplexity does not move.

8 Conclusion

The encoder reaches strong length-category accuracy and reasonable sentiment performance (macro F1 0.54). Retrieval at $k = 3$ gives the best match rate and provides context for explanation generation. The decoder learns the explanation template well, leading to very low perplexity. The main limitation is the template target; future work should add more natural and diverse explanation targets to test RAG gains on richer language.